Dietary meat and fat intake and prevalence of rhinoconjunctivitis in pregnant Japanese women: baseline data from the Kyushu Okinawa Maternal and Child Health Study

Background Dietary fat exerts numerous complex effects on proinflammatory and immunologic pathways. Several epidemiological studies have examined the relationships between intake of fatty acids and/or foods high in fat and allergic rhinitis, but have provided conflicting findings. The current cross-sectional study investigated such relationships in Japan. Methods Study subjects were 1745 pregnant women. The definition of rhinoconjunctivitis was based on criteria from the International Study of Asthma and Allergies in Childhood. Information on dietary factors was collected using a validated self-administered diet history questionnaire. Adjustment was made for age; gestation; region of residence; number of older siblings; number of children; smoking; secondhand smoke exposure at home and at work; family history of asthma, atopic eczema, and allergic rhinitis; household income; education; and body mass index. Results The prevalence of rhinoconjunctivitis in the past 12 months was 25.9%. Higher meat intake was significantly associated with an increased prevalence of rhinoconjunctivitis: the adjusted odds ratio between extreme quartiles was 1.71 (95% confidence interval: 1.25-2.35, P for trend = 0.002). No measurable association was found between fish intake and rhinoconjunctivitis. Intake of total fat, saturated fatty acids, monounsaturated fatty acids, n-3 polyunsaturated fatty acids, α-linolenic acid, eicosapentaenoic acid, docosahexaenoic acid, n-6 polyunsaturated fatty acids, linoleic acid, arachidonic acid, and cholesterol and the ratio of n-3 to n-6 polyunsaturated fatty acid intake were not evidently related to the prevalence of rhinoconjunctivitis. Conclusions The current results suggest that meat intake may be positively associated with the prevalence of rhinoconjunctivitis in young adult Japanese women.